When Does Density Beat Centroids? A Comparative Study of Clustering Methods Across Data Geometries

[**DECIDE**: final title. Proposed: "When Clustering Fails: Which Data Break Which Methods, and Can You Tell Without Labels?", to match the research question]

SC4020 Project 1 — Group XX

[**All**: full names exactly as on student cards; replace XX with group ID]

Abstract

Clustering is used when there are no labels to check the result against, so a clustering is usually judged by an internal score such as the silhouette. We test whether that is safe. The methods assume different things: K-means expects compact groups, while DBSCAN and HDBSCAN expect dense groups separated by empty space. We ran K-means (with random and K-means++ starts), DBSCAN and HDBSCAN on eight datasets chosen to break these assumptions. Six of the datasets have known labels. We asked three questions: does a method fail when its assumption breaks, can the failure be seen without labels, and does a known remedy fix it? Every method failed where its assumption broke. K-means cut two curved clusters in the wrong place (agreement with the true labels 0.48 out of 1, against 0.98 for DBSCAN), and on housing data with no empty space between groups both density methods put most points in one cluster. The silhouette did not reveal these failures: on five of the six labelled datasets it favoured a different method from the one that best matched the labels. Comparing each result with the same method run on shuffled data caught results with no real structure, but not K-means joining earthquake belts thousands of kilometres apart into one cluster. Remedies such as measuring distance on the sphere and using HDBSCAN fixed the failures they target, but none helped data with no gaps. No single score can choose a clustering method: an analyst should look at the data, report how many points each method discards, and test directly for the failures the data make likely.

1 Introduction

1.1 Motivation

Clustering is used where no answer key exists. A seismologist grouping earthquake epicentres wants active fault systems; a housing analyst grouping resale transactions wants market segments. Neither has labels to check a result against.

Each clustering method carries its own idea of what a cluster is. K-means [12] looks for compact groups around a centre and assigns every point to one of them. DBSCAN [8] and HDBSCAN [5] look for dense regions separated by sparse space, and leave points in the sparse space as noise. Whether or not that idea matches the data, the method returns a partition. On one year of global earthquakes, K-means returns six tidy regions, one of which joins five plate boundaries lying 2,932 to 11,449 km apart (Section 3.4.1). Nothing in the output says that this has happened. An analyst without labels usually judges the result by an internal score such as the silhouette [17], which rates how compact and well separated the clusters are, and that score is itself built on the K-means idea of a cluster.

1.2 Research question

This report asks one question: *when a clustering method's idea of a cluster does not match the data, does it fail, and can the failure be detected without labels?* We split it into three parts.

RQ1, cause. Which property of the data breaks which method? We consider cluster shape, density that varies between clusters, the number of dimensions, the absence of gaps between groups, feature scale, and values recorded in steps.

RQ2, detection. Do checks that need no labels reveal those failures? We test the silhouette, the Davies–Bouldin index, the share of points labelled noise, and a fake-data check: the same method run on data whose features have been shuffled independently, which keeps each feature's values but removes any joint structure.

RQ3, remedy. For each failure, does a known remedy help, and what does it cost?

We compare four methods, K-means with random starts, K-means++ [2], DBSCAN and HDBSCAN, on eight datasets (Table 2). Each dataset was chosen to test one data property, most of them a property that breaks at least one method's assumption. Six have class labels, so a method's output can be scored against them with the Adjusted Rand Index (ARI) [11], and the internal scores can be checked against that answer. Two have no labels (earthquakes, Uber pickups); there we judge failures on maps, against known geography. Gaussian mixtures, spectral clustering and agglomerative clustering appear only as reference points on the labelled datasets where we ran them (D1, D5, D6, D7), not as part of the main comparison.

1.3 Findings in brief

RQ1. Each method failed where the data broke its assumption. K-means split the two interleaved half-moons of D1 wrongly (ARI 0.48) and merged distant belts on D2. With one radius, DBSCAN clustered 90% of earthquakes in the dense south-west Pacific but only 20% in the sparser Americas. In 50 dimensions (D5) it barely matched the labels (ARI 0.03). On D8, where housing sales form one mass with no gaps, both density methods returned one dominant cluster.

RQ2. The silhouette did not reveal these failures. On five of the six labelled datasets, the method with the highest silhouette was not the one that matched the labels best. On D1 it ranks K-means first (0.495 against 0.392 for DBSCAN), although DBSCAN recovers the moons (ARI 0.98 against 0.48). On Ecoli (D4) it orders the four methods almost exactly opposite to ARI (Spearman $\rho = -0.80$). The fake-data check, which we ran on D2, D3, D4 and D8, does better, but only partly. It detects when a method finds nothing real: on D8 both density methods score below the same setting on shuffled data, and on D4 the HDBSCAN setting chosen without labels scores no better than on shuffled data (0.559 against 0.551). It does not detect the wrong structure: on D2, K-means scores well above shuffled coordinates (silhouette 0.528 against 0.391) while merging unrelated belts. Two failures were found only by checks aimed at a known problem: clusters cut at the 180° meridian, and clusters that are just one recorded storey band.

RQ3. Known remedies repaired some failures and not others. Measuring distance on the sphere kept all 92 close earthquake pairs across the 180° meridian together, where flat coordinates kept none. HDBSCAN raised the clustered share of the Americas from 20% to 62%, at the cost of trimming the densest belts (78% against 90%). Nothing we tried produced meaningful density clusters on D8 with all four features: capping the size of the largest cluster only made the rule select storey bands.

Section 2 describes the methods and Section 3 the datasets, protocol and results, grouped by data property. Section 4 turns them into guidance.

1.4 Related work

Centroid and mixture models. Lloyd's algorithm [12] minimises the within-cluster sum of squared distances by alternating assignment and centroid updates. It converges to a local optimum

that depends on the starting centres. K-means++ [2] picks each starting centre with probability proportional to its squared distance from the centres already chosen, which gives an $O(\log k)$ approximation to the optimum in expectation. A Gaussian mixture fitted by EM [7] generalises K-means to ellipsoidal clusters with soft assignments. Agglomerative clustering builds a hierarchy by repeatedly merging the closest pair of clusters. Under Ward's criterion [19] the merge cost is the increase in within-cluster variance, the same objective as K-means.

Density-based methods. DBSCAN [8] defines a cluster as a maximal set of density-connected points. A point is a core point if at least `minPts` points lie within radius ε. Points reachable from no core point are noise. The method needs no k and imposes no shape, but its single global ε assumes all clusters have similar density. OPTICS [1] orders points by reachability so that clusters at several densities can be read from one plot. HDBSCAN [5] builds a hierarchy over mutual reachability distances and keeps the clusters that persist longest. This removes ε and leaves a minimum cluster size as the main parameter.

Graph-based methods. Spectral clustering [15] embeds the points using the leading eigenvectors of a normalised affinity matrix, then runs K-means in that embedding. Groups connected through the affinity graph become compact in the embedding, so non-convex clusters can be separated. A dense affinity matrix needs $O(n^2)$ memory, which limits the method to modest n.

Validation and cluster tendency. Internal indices, such as the silhouette [17], Davies–Bouldin [6] and Calinski–Harabasz [4] indices, score a partition from the data alone. External indices such as the Adjusted Rand Index [11] compare it with known labels. Von Luxburg et al. [13] argue that no criterion can judge a clustering apart from its intended use. A separate line of work tests whether data contain clusters at all. The Hopkins statistic [9] compares nearest-neighbour distances in the data with those in uniformly random points. The gap statistic [18] compares within-cluster dispersion with its value on reference data drawn from a null distribution. Our fake-data check applies the same idea to every method we compare. In high dimensions the ratio of nearest to farthest neighbour distance tends to one [3], which weakens any method built on a density threshold.

2 Methods

2.1 K-Means and K-Means++

K-means [12] looks for k centres $\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_k$ that minimise the within-cluster sum of squares (inertia), $J = \sum_{i=1}^{n} \min_j \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2$. Each iteration assigns every point to its nearest centre and moves each centre to the mean of its points. J never rises, so the method stops at a local optimum that depends on the starting centres. One iteration costs $O(nkd)$. Random starts pick k data points uniformly. K-means++ [2] picks the first centre at random and each next one with probability proportional to its squared distance from the nearest centre already chosen. Only the start differs, so we treat the two as one method with two initialisations (Ablation A). We use scikit-learn's `KMeans` with 10 starts; the only parameter is k. Because J sums squared Euclidean distances, K-means suits compact, roughly round clusters of similar spread. It cuts curved or elongated clusters, gives every point a cluster, so outliers pull the centres, and depends on the scale of each feature (Ablation C).

2.2 DBSCAN

DBSCAN [8] takes a radius ε and a count `minPts`. A point is a core point if at least `minPts` points lie within ε of it. Core points within ε of each other join the same cluster, points within ε of a core point join its cluster as border points, and every other point is noise. The method needs no k

and assumes no shape, and on coordinates ε is a distance on the ground that an analyst can judge. With a spatial index a fit costs about $O(n \log n)$ in few dimensions, rising towards $O(n^2)$ in many dimensions or with a large ε. A reference value for ε comes from the k-distance curve, each point's distance to its `minPts`-th nearest neighbour in sorted order; its knee is the point farthest from the line joining the two ends of the curve. The weakness is the single radius. It assumes all clusters have similar density, so a radius that separates dense clusters calls sparse ones noise, and one that keeps sparse clusters merges dense ones. In many dimensions distances become alike [3], and a radius separates little.

2.3 HDBSCAN

HDBSCAN [5] removes ε. Each point's core distance is the distance to its m-th nearest neighbour, and the mutual reachability distance $d_{\mathrm{mr}}(a, b) = \max\{\mathrm{core}(a), \mathrm{core}(b), d(a, b)\}$ pushes points in sparse regions apart. A minimum spanning tree under d_{mr} holds the DBSCAN clusterings at every density level at once. The tree is condensed by dropping any split that leaves fewer than `min_cluster_size` points, and the clusters that persist over the widest range of density are kept; the remaining points are noise. Different branches can thus be cut at different densities, which is what DBSCAN cannot do. We use scikit-learn's `HDBSCAN`; the main parameter is `min_cluster_size`, and m (`min_samples`) defaults to it. A fit costs between $O(n \log n)$ and $O(n^2)$. Because `min_cluster_size` is a count, the result changes with the size of the sample. HDBSCAN can call large parts of the data noise, and like DBSCAN it has nothing to find where the data have no density gaps.

2.4 Gaussian Mixture Model

A Gaussian mixture model (GMM) treats the data as drawn from k Gaussians, $p(\mathbf{x}) = \sum_{j=1}^{k} \pi_j \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$, and fits the weights π_j, means and covariances by expectation–maximisation (EM) [7]. The E-step computes each point's responsibility (posterior probability) under every component; the M-step re-estimates π_j, $\boldsymbol{\mu}_j$ and $\boldsymbol{\Sigma}_j$ from those soft assignments; the two steps repeat until the log-likelihood stops improving. Each point is then given its most probable component. We use scikit-learn's `GaussianMixture` [16] with *full* covariance matrices (the default), k-means initialisation, one start and a fixed seed. The only parameter is the number of components k, which we set to the number of classes on labelled data and to the silhouette-best $k \in [2, 10]$ otherwise. One EM iteration costs $O(nkd^2 + kd^3)$. GMM generalises K-Means: full covariances allow ellipsoidal clusters of different size, orientation and weight, but each cluster must still be a single convex blob. It fails on curved or nested shapes, has no noise label, and with full covariances it needs many points per component when d is large.

2.5 Spectral clustering

Spectral clustering [15] turns clustering into graph partitioning. It (i) builds an affinity matrix W whose entry w_{ij} measures how similar points i and j are; (ii) forms the normalised graph Laplacian and takes its k leading eigenvectors, giving each point a k-dimensional embedding; (iii) runs K-Means on the rows of that embedding. Points joined by a chain of close neighbours land near each other in the embedding even when they are far apart in the input space, so the method can recover curved and non-convex clusters. We use scikit-learn's `SpectralClustering` with two affinities: an RBF kernel $w_{ij} = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2)$, $\gamma = 1$, for roughly globular data (D6 Wine, Mall), and a symmetric 10-nearest-neighbour graph for manifold-shaped data (D1 make_moons, D5 Fashion-MNIST), where one global bandwidth fits poorly. Parameters are k and the affinity (with γ or the neighbour count). A dense RBF affinity costs $O(n^2 d)$ time and $O(n^2)$ memory, and a full eigendecomposition $O(n^3)$; the sparse k-NN graph with an iterative eigensolver is much cheaper but still limits n (Section 3.7). Spectral clustering needs k, has no noise label, and is sensitive to the affinity scale.

2.6 Agglomerative clustering

Agglomerative clustering starts with every point as its own cluster and repeatedly merges the two closest clusters until k remain. The linkage defines which two are closest. We use Ward linkage [19], which merges the pair whose union raises the total within-cluster variance least. This is the K-means objective, searched greedily from the bottom up. The method therefore needs no starting centres and returns the same partition on every run, which makes it a direct check on K-means: same objective, different search. The only parameter is k, set to the number of classes (3 on D6, 2 on D7); distances are Euclidean on standardised features. It needs $O(n^2)$ time and memory for the pairwise distances, so we ran it on the two small tables only ($n = 178$ and $n = 569$). Like K-means, it assumes compact clusters, assigns every point and has no noise label. Its own weakness is that a merge is never undone, so an early mistake stays in the final partition. On both datasets it matched the labels less well than K-means (ARI 0.790 against 0.897 on D6, 0.575 against 0.654 on D7).

Table 1: Structural assumptions of the methods compared.

	K-Means	K-Means++	DBSCAN	HDBSCAN	GMM	Spectral	Agglom.
Cluster shape assumed	isotropic	isotropic	arbitrary	arbitrary	ellipsoidal	connected (graph)	compact (Ward)
Requires k	yes	yes	no	no	yes	yes	yes
Explicit noise label	no	no	yes	yes	no	no	no
Handles varying density	no	no	no	yes	partly[b]	partly[c]	no
Parameter is interpretable	no	no	yes (km)	partly	no	no	no
Time complexity	$O(nkdi)$	$O(nkdi)$	$O(n \log n)$[a]	$O(n \log n)$[a]	$O(nkd^2i)$	$O(n^2d + n^3)$[d]	$O(n^2)$

[a] With an effective spatial index; $O(n^2)$ worst case. [b] Per-component covariance and weight. [c] With a k-NN affinity, which adapts to local scale. [d] Dense RBF affinity and full eigendecomposition; lower with a sparse k-NN graph.

3 Experiments

3.1 Datasets

Table 2: Datasets, grouped by the data property each one tests. [**DECIDE**: final list; drop or keep Mall; Ky's 1965–2016 earthquakes replaced by the 2023 surface-distance pipeline?]

ID	Dataset	n	d	Labels	Property tested
D1	make_moons (synthetic)	1,000	2	2 classes	curved shapes (control)
D2	Earthquakes, 2023, M≥4.5	7,643	2	none	long belts, uneven density
D3	Uber NYC pickups, Apr 2014	15,000	2	none	density contrast in a city
D4	Ecoli	336	7	8 classes	dimensionality, rare classes
D5	Fashion-MNIST (PCA)	3,000	50	10 classes	high dimensions
D6	Wine	178	13	3 classes	silhouette vs labels
D7	Breast cancer (Wisconsin)	569	30	2 classes	silhouette vs labels
D8	HDB resale flats, 2017–2026	20,000[b]	4	flat type (7)	no gaps, scale, recorded ties

[b] Random sample of 234,294 usable sales; all 234,294 used for runtime (Section 3.7).

D1 make_moons. Two interleaving half-circles from scikit-learn's `make_moons` [16]: 1,000 points, Gaussian noise 0.07, seed 42, then standardised. The two classes are dense, equal in size and clearly separated by a gap, but neither is convex: each moon wraps around the end of the other, so the straight boundary a centroid method draws must cut both. With the true labels known and no other difficulty present, D1 isolates cluster shape as the single property under test.

D2 Earthquakes. Every event of magnitude 4.5 or more in 2023 from the USGS ComCat catalogue [22]: 7,651 events, of which 8 volcanic eruptions near Saipan are removed, leaving 7,643

earthquakes. We cluster the epicentre only. Depth is left out because 44.2% of events carry the default depth of 10 km. Each method meets a different difficulty. Earthquakes lie along narrow belts thousands of kilometres long, which breaks the compact-cluster assumption of K-means. Density varies widely: the distance to the nearest earthquake runs from 1.2 km (5th percentile) to 100.5 km (95th), so no single DBSCAN radius suits every belt. And 17.2% of earthquakes lie within 10° of the 180° meridian, where flat coordinates cut the map. There are no labels.

D3 Uber NYC. Uber pickups in New York City in April 2014, from the trip records the city's Taxi and Limousine Commission released to FiveThirtyEight [27]: 564,516 pickups, each a latitude and longitude. We keep the 558,019 inside a box around the city (latitude 40.57 to 40.92, longitude −74.15 to −73.70), which drops GPS errors, and sample 15,000 at random (seed 42). For K-means and DBSCAN the coordinates are projected to kilometres about the mean latitude, so ε is a distance on the ground. Pickups crowd into lower and midtown Manhattan (71% of the sample), with smaller dense spots at the two airports and sparse demand in the outer boroughs. The distance to the 8th nearest pickup runs from 0.017 km to 0.418 km (5th to 95th percentile), but the dense and sparse areas meet with no clear gap between them. There are no labels.

D4 Ecoli. The UCI Ecoli data [26]: 336 proteins of *E. coli*, each described by seven scores computed from its amino-acid sequence and labelled with one of eight sites in the cell where the protein is found. Features are standardised. The classes are very unequal: the five largest hold 143, 77, 52, 35 and 20 proteins, the three smallest 5, 2 and 2. D4 is hard for density methods in two ways. Seven dimensions already spread the points so that a density threshold separates less well, and the rare classes are too small to form a dense region of their own, so a density method tends to call them noise.

D5 Fashion-MNIST. Fashion-MNIST [20] holds 70,000 28×28 greyscale images of 10 clothing classes. We pool the train and test files, scale pixels to $[0, 1]$, draw a class-stratified sample of 3,000 images (300 per class, seed 42), and project the 784 pixels onto 50 principal components, which keep 86.5% of the variance. The sample keeps Spectral, DBSCAN and HDBSCAN within reach on a laptop (Section 3.7) while keeping the classes balanced. D5 is hard because the space is high dimensional, so distances concentrate [3], and because several classes (shirt, T-shirt, pullover, coat) overlap heavily with no density gap between them.

D6 Wine. The UCI Wine data [24]: 178 wines grown in one Italian region, each described by 13 chemical measurements such as alcohol, malic acid, flavanoids and proline. The cultivar (3 classes of 59, 71 and 48 wines) is held out as the label. Features are standardised, because proline runs from 278 to 1,680 while hue stays below 2. The classes are compact, so K-means recovers them well: two members ran it independently and both obtained ARI 0.897. The dataset is in the study for the opposite reason. No sparse gap separates the cultivars, so the density methods return 2 clusters with about a third or more of the wines as noise, and the silhouette still prefers them (Section 3.4.5).

D7 Breast cancer. The Wisconsin diagnostic breast cancer data [25]: 569 tumours, each described by 30 measurements of the cell nuclei in an image of a fine-needle aspirate (the mean, standard error and largest value of 10 properties such as radius, texture and concavity). The diagnosis, 357 benign and 212 malignant, is held out as the label. Features are standardised, because areas reach 4,254 while smoothness and symmetry stay below 1. Two properties make the data hard. The classes differ in spread: malignant tumours lie 35% further from their class centre than benign ones (5.47 against 4.06 in standardised units), which breaks the equal-size assumption of K-means. And the two classes form one connected mass in 30 dimensions, with no sparse gap for a density method to cut along.

D8 HDB resale. Every resale of a public housing flat in Singapore from January 2017, from data.gov.sg [23]. The 6,051 sales after June 2026 (2.5%) have no price index value yet and are dropped, leaving 234,294. Methods are compared on a random sample of 20,000 (seed 0); all 234,294 are used for runtime. There are four features: floor area, remaining lease, storey, and price adjusted to 2026 levels with the HDB resale price index. Flat type (7 classes) is held out as a label, but it is nearly a function of size: floor area alone reaches ARI 0.601 against it. The data are hard for density methods. There are no gaps between groups of flats, and storeys are recorded in 3-storey bands, so 77.7% of sales share their floor area, lease and storey with at least one other sale.

3.2 Preprocessing

Three rules apply wherever a dataset has the property they address. Each is tested by an ablation in Section 3.5.

Coordinates on the sphere. Latitude and longitude treated as flat numbers put two earthquakes 6.07 km apart at 359.94° of longitude from each other, because they lie on either side of the 180° meridian. Density methods therefore use the haversine (great-circle) distance on coordinates in radians. K-means needs a vector space to average in, so each epicentre becomes a 3D point on the unit sphere. The straight-line distance between two such points grows with their great-circle distance, so the clustering follows surface distance (Ablation B).

Standardised features. Features in their raw units are not comparable. On D8, price (in dollars) accounts for 100.0% of the squared distance between sales, so the other three features are ignored. Every feature is scaled to mean 0 and standard deviation 1 (Ablation C).

Values recorded in steps. When a feature is recorded in steps, many points share exact values, and density methods read each shared value as a dense spot. Before running a density method on D8, each stepped value is moved by a uniform random amount of up to half its step: ± 1.5 storeys, $\pm 0.5\,\mathrm{m}^2$ of floor area and $\pm\frac{1}{24}$ year of lease. Price is continuous and is left unchanged. Results are reported with and without this jitter (Ablation E).

3.3 Evaluation metrics and parameter selection

Metrics. Every result table reports the same columns. *Internal* scores need no labels. The silhouette coefficient [17] of a point is $(b-a)/\max(a,b)$, where a is its mean distance to its own cluster and b its mean distance to the nearest other cluster; we report the mean over points, from -1 to 1, higher is better. The Davies–Bouldin index [6] averages, over clusters, the worst ratio of within-cluster spread to between-centre distance; lower is better. Both are computed on clustered points only, so they say nothing about the points a density method calls noise. Two *coverage* figures therefore stand next to them: the noise fraction and the share of points in the largest cluster. *External* scores need labels. The Adjusted Rand Index (ARI) [11] counts pairs of points on which a clustering and the labels agree, corrected so that a random clustering scores about 0 and a perfect one 1. Normalised mutual information (NMI) measures the same agreement in terms of shared information. For both, noise counts as one more cluster, so a method cannot raise its score by discarding hard points. We also report the fit time.

Fake-data check. An internal score has no fixed scale: a silhouette of 0.3 can be strong on one dataset and meaningless on another. Each chosen setting is therefore run again on fake data that keeps each feature's distribution but has no joint structure. For tables, each column is shuffled independently. For coordinates, points are drawn uniformly in the same latitude and longitude range. A result counts as structure only if the real data score clearly above the fake data.

Selection rule. One rule picks the setting of DBSCAN and HDBSCAN on every dataset: among the settings that give at least 2 clusters and label at most 50% of points as noise, take the one with the highest silhouette. The noise cap is needed because the silhouette ignores noise. Without it, the silhouette prefers settings that throw most of the data away. On D2 it would pick DBSCAN with $\varepsilon = 20.4$ km, which scores 0.878 by calling 69.4% of earthquakes noise. On D8 it would pick HDBSCAN with `min_cluster_size` 400, which scores 0.448 with 70.9% of sales as noise. For K-means, K is chosen from the elbow in inertia, the silhouette, the gap to the fake-data silhouette and agreement between seeds; each dataset states which of these decided. On labelled data we report the rule's choice and the setting with the best ARI, so the cost of choosing without labels is visible.

Parameter grids. K-means and K-means++: $K = 2$ to 12, 10 starts each. DBSCAN: `minPts` (`min_samples` in scikit-learn) in $\{5, 10, 20\}$. Its radius ε comes from the k-distance curve, the sorted distance from each point to its `minPts`-th neighbour: on D2 we take the 25th, 50th, 75th, 90th and 95th percentiles of the curve (11 to 810 km); on D8 we take 0.5, 0.75, 1, 1.5 and 2 times the knee of the curve. HDBSCAN: `min_cluster_size` scaled to the data (25, 50, 100 on D2; 100, 200, 400 on D8) and `min_samples` in {default, 5, 10}. On D6 and D7, K-means is searched over $K = 2$ to 10 with 10 starts and reported at the number of classes. GMM takes 2 to 10 components and four covariance types (full, tied, diagonal, spherical), chosen by BIC. DBSCAN takes `minPts` in $\{10, 20, 40, 80\}$ and ε at 0.5 to 4 times the knee of the k-distance curve. HDBSCAN takes `min_cluster_size` from 5 up to $n/10$ (15 on D6, 50 on D7) and `min_samples` in {default, 5, 10}. On these two datasets the density settings that pass the constraints (at least 2 clusters, at most 50% noise, and at most 10 clusters) are ranked by DBCV [14], a score built for density clusters, instead of the silhouette. Agglomerative clustering uses Ward linkage at the number of classes. GMM and Spectral on D1 and D5: k set to the number of classes on labelled data, else the silhouette-best K-Means++ $k \in [2, 10]$; GMM full covariance; spectral affinity RBF ($\gamma = 1$) for D6 and Mall, 10-nearest-neighbour graph for D1 and D5. DBSCAN and HDBSCAN on D1 and D5: `minPts` from $\{\lfloor m/2 \rfloor, m, 2m\}$ with $m = \text{clip}(2d, 4, 20)$, ε from $\{0.5, 0.75, 1, 1.5, 2\}\times$ the k-distance knee; `min_cluster_size` from $\{5, 10, 20, 30, 50, n/20\}$; the setting with the best silhouette (noise excluded, ≥ 2 clusters, $\leq 50\%$ noise) is kept. On D3 and D4: K-means and K-means++ $K = 2$ to 25, 10 starts. DBSCAN `minPts` in $\{4, 8, 16, 32\}$ (D3) or $\{3, 5, 8, 12\}$ (D4), with ε at 0.1 to 3 times the knee of the k-distance curve. HDBSCAN `min_cluster_size` from 10 to 250 (D3) or 5 to 30 (D4). The shared rule picks the density settings. On D3 the silhouette alone picks $K = 2$; we use $K = 5$, where the gap to fake data is larger (0.113 against 0.089). On D4 the silhouette picks $K = 5$, and $K = 7$ gives the best ARI.

Reproducibility. On D2 and D8 every random step uses seed 0, and repeated runs use seeds 0 to 19. They ran on Python 3.12.13, scikit-learn [16] 1.9.1 and NumPy 2.5.3 on an Apple M2 with 16 GB of memory. On D3 and D4 the same library versions ran on Python 3.12.10, on an Intel Core Ultra 5 225 with 32 GB of memory; every random step uses seed 42, and repeated runs use seeds 0 to 29. D1 and D5 ran on Python 3.13.15 with scikit-learn 1.9.1 and NumPy 2.5.3. D6 and D7 ran on Python 3.13.7 with scikit-learn 1.9; their models use seed 42 and the column shuffle for the fake-data check uses seed 0. Each dataset has its own code, but every pipeline computes the scores with the same scikit-learn functions, so the tables in Section 3.4 share their columns. [**Lam**: model seed for D1 and D5, and hardware (CPU, memory).] [**Hoang Anh**: hardware (CPU, memory) and exact scikit-learn version for D6 and D7.]

3.4 Comparison by data property

[**All**: One results table per property (rows: dataset × method; columns: #clusters, noise, S, DB, ARI, NMI, time, fake-data gap). Bold the best per dataset.]

Table 3: D2 and D8 at the settings chosen in Section 3.3. S: silhouette; DB: Davies–Bouldin; null S: silhouette of the same setting on fake data (D2: shuffled coordinates for K-means, uniform points for density methods; D8: shuffled columns). "none": no clusters formed on the fake data. D2 has no labels.

Data	Method	Clusters	Noise	Largest	S	DB	ARI	NMI	Time (s)	Null S
D2	K-Means ($K = 6$)	6	0%	44.3%	0.525	0.800	–	–	0.004	0.389
D2	K-Means++ ($K = 6$)	6	0%	31.0%	0.528	0.738	–	–	0.005	0.391
D2	DBSCAN (101.8 km, 20)	41	40.4%	11.8%	0.703	0.351	–	–	0.051	none
D2	HDBSCAN (50)	28	30.7%	7.5%	0.757	0.329	–	–	1.081	0.320
D8	K-Means ($K = 4$)	4	0%	36.2%	0.317	1.116	0.220	0.299	0.011	0.195
D8	K-Means++ ($K = 4$)	4	0%	35.1%	0.317	1.116	0.213	0.293	0.010	0.218
D8	DBSCAN (0.567, 10)	3	1.4%	97.1%	0.239	0.732	0.030	0.091	0.428	0.330
D8	HDBSCAN (200)	2	28.5%	68.6%	0.160	1.133	0.072	0.101	2.022	0.593

Table 4: D3 and D4, same columns as Table 3. Top: settings chosen without labels (Section 3.3). Bottom: D4 settings with the best ARI, which need the labels. "none": fewer than 2 clusters formed on the fake data. D3 has no labels.

Data	Method	Clusters	Noise	Largest	S	DB	ARI	NMI	Time (s)	Null S
D3	K-Means ($K = 5$)	5	0%	44.5%	0.475	0.699	–	–	0.064	0.361
D3	K-Means++ ($K = 5$)	5	0%	44.5%	0.475	0.699	–	–	0.036	0.362
D3	DBSCAN (1.58 km, 32)	2	1.9%	96.1%	**0.779**	0.172	–	–	0.521	none
D3	HDBSCAN (250)	3	4.8%	91.2%	0.641	0.266	–	–	1.111	0.395
D4	K-Means ($K = 5$)	5	0%	44.6%	0.408	0.743	0.684	0.652	0.022	0.191
D4	K-Means++ ($K = 5$)	5	0%	44.0%	0.405	0.747	**0.697**	0.666	0.015	0.185
D4	DBSCAN (2.09, 5)	2	1.8%	96.7%	**0.560**	0.504	0.039	0.135	0.002	0.548
D4	HDBSCAN (5)	2	0.3%	97.0%	0.559	0.637	0.038	0.118	0.002	0.551
D4	K-Means++ ($K = 7$), best ARI	7	0%	43.5%	0.354	0.861	**0.731**	0.679	0.017	0.195
D4	DBSCAN (0.97, 8), best ARI	2	21.1%	53.6%	0.434	0.908	0.489	0.488	0.001	0.278
D4	HDBSCAN (12), best ARI	2	37.2%	44.3%	0.483	0.789	0.339	0.376	0.002	none

3.4.1 Curved and elongated clusters

On D1 the centroid methods fail and the graph and density methods succeed. K-Means++ (ARI 0.481) and GMM (0.489) both draw a straight boundary that cuts each moon in two, giving the tip of one moon to the other cluster (Figure 10). Spectral clustering with a 10-nearest-neighbour graph recovers both moons exactly (ARI 1.000), DBSCAN nearly so (0.982, 0.9% noise) and HDBSCAN slightly less (0.947, 2.7% noise, mostly the sparse moon tips). Silhouette ranks the methods the other way round: K-Means++ and GMM score 0.495 against 0.387 for Spectral, because silhouette rewards compact, convex groups, which is exactly the wrong assumption here.

K-means assumes compact clusters, and D2 breaks that assumption at planetary scale. Earthquakes follow plate boundaries, so a cluster is a long, narrow belt rather than a ball around a centre. K-Means++ with $K = 6$ must cover the whole globe with six roughly round regions, so each region takes in several unrelated belts (Figure 1). One cluster of 905 earthquakes joins Turkey, the Reykjanes Ridge and Iceland, Afghanistan and Tajikistan, Somalia and Eritrea, and the Mid-Indian Ridge. HDBSCAN finds each of these as its own cluster, and their centres are 2,932 to 11,449 km apart. K-means also has no noise label, so every isolated earthquake in the oceans is assigned to some region. Its silhouette is still 0.528, well above the 0.391 of shuffled coordinates: the score confirms that the data are not random, not that the clusters are meaningful.

3.4.2 Uneven density

On D2 the distance to the nearest earthquake runs from 1.2 km to 100.5 km (5th to 95th percentile), and the k-distance curve rises smoothly with no knee. Any single ε is therefore too small for some belts and too large for others. At the chosen $\varepsilon = 101.8$ km, DBSCAN clusters 90% of the earthquakes in the dense south-west Pacific but only 20% in the sparser Americas (Figure 2).

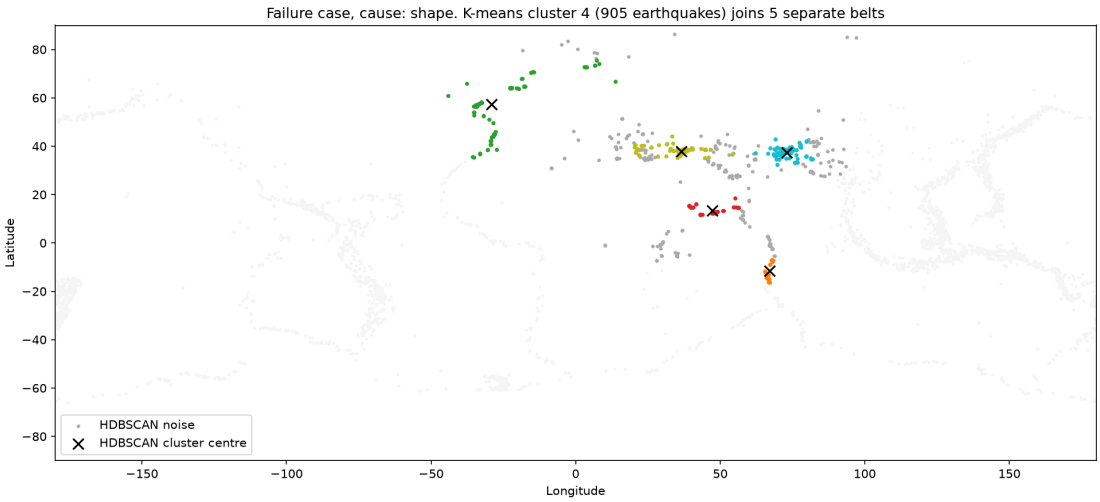


Figure 1: D2, failure of K-Means++ ($K = 6$). All points shown in colour or dark grey belong to one K-means cluster; colours are the HDBSCAN clusters inside it, crosses their centres. Cause: cluster shape. Belts thousands of kilometres apart are joined because K-means can only draw compact regions.

HDBSCAN, which picks a density level per cluster, clusters 62% of the Americas and keeps the Mexico to Panama belt and the Chile and Argentina belt as long clusters. The price is paid in the dense region, where it clusters 78% instead of 90%. Overall HDBSCAN keeps more earthquakes (69.3% against 59.6%) with a higher silhouette (0.757 against 0.703, Table 3).

On D3 density varies as much, but inside one connected mass. Lower and midtown Manhattan, the rest of Manhattan, Brooklyn and western Queens meet without a gap, so the density methods keep them together. DBSCAN at the knee of the k-distance curve ($\varepsilon = 0.525$ km) returns 13 clusters, yet 91.6% of pickups sit in the first; the others are the two airports and small spots in Jersey City, Astoria, Harlem and Brooklyn, and the silhouette is 0.04. The rule prefers $\varepsilon = 1.58$ km: two clusters, the city (96.1%) and JFK, with a silhouette of 0.779 (Table 4). HDBSCAN finds the city, JFK and LaGuardia (91.2%, 2.0% and 1.9%; 4.8% noise). The airports are the only parts of D3 with a gap around them, and both density methods find them. K-means ($K = 5$) does divide the core, into southern Manhattan (44.5%), northern Manhattan with the Bronx (39.9%) and Brooklyn (9.3%), plus the two airport areas, but those borders follow the K-means objective, not gaps in the data. Its silhouette is 0.475 against 0.362 on shuffled coordinates, so the split is not random, but nothing in the data marks where it should fall.

3.4.3 High dimensions

D4 has only seven dimensions, but it already shows the effect. Given its best ε in each case, DBSCAN beats K-Means++ on the first two and three principal components (ARI 0.575 and 0.618 against 0.468 and 0.466) and loses from four components on, by 0.276 on all seven (0.455 against 0.731; Ablation D). Without labels the density methods do far worse. On all seven features the rule picks settings where one cluster holds 97% of proteins, with ARI 0.04 (Table 4), and their silhouettes (0.560 and 0.559) are barely above the same settings on shuffled columns (0.548 and 0.551). The fake-data check flags both. K-means at the $K = 5$ the silhouette picks reaches ARI 0.684 to 0.697. On D5 ($d = 50$) the ranking reverses from D1. The k-based methods do best: Spectral (ARI 0.398), GMM (0.391) and K-Means++ (0.343), all well short of the labels because shirts, T-shirts, pullovers and coats overlap. DBSCAN finds two clusters and 33% noise (ARI 0.033): one cluster absorbs almost every class (Figure 13). HDBSCAN finds 28 small clusters but labels 71% of points noise (ARI 0.050). No setting in the HDBSCAN grid met the 50% noise limit, because in 50 dimensions the k-distances of all points are similar and there is no density gap between the overlapping classes for one threshold to find.

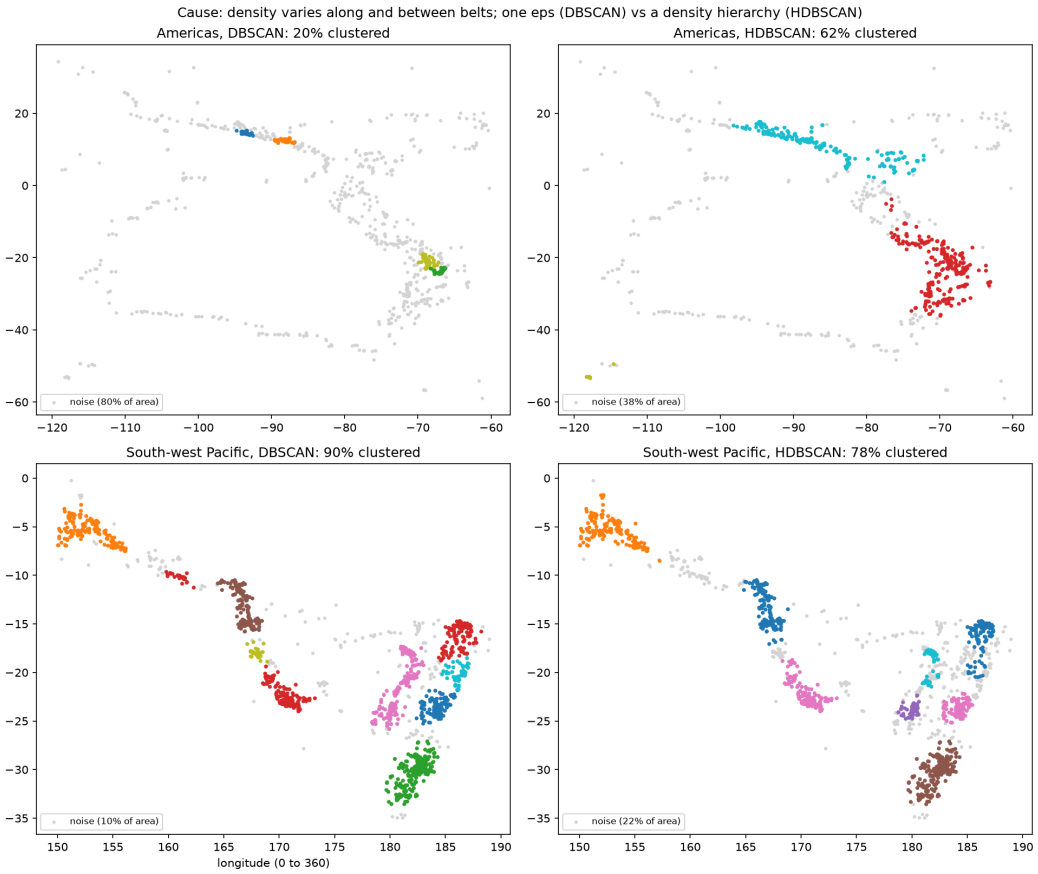


Figure 2: D2, DBSCAN ($\varepsilon = 101.8\,\text{km}$) against HDBSCAN in a sparse region (top, the Americas) and a dense one (bottom, the south-west Pacific); grey points are noise. Success of HDBSCAN on the sparse belts, at the cost of trimming the dense ones. Cause: density varies along and between belts.

3.4.4 No gaps between groups

D8 is one continuous mass: flats come in every combination of size, lease, storey and price, with no empty space between groups. Table 3 shows what each method does with it. K-means still finds a weak partition, four market segments (newer mid-size, older small, large older, newer high-floor), with a silhouette of 0.317 against 0.218 on shuffled columns. That ratio of 1.45 is modest, but it is the only result on D8 that beats its fake data. Density methods have no sparse region to cut along. The DBSCAN setting chosen by the rule puts 97.1% of sales in one cluster, and HDBSCAN finds one core holding 68.6% of sales plus a compact block of new 3-room flats, and calls 28.5% noise. Both score below the same setting on shuffled columns (0.239 against 0.330, and 0.160 against 0.593). On shuffled columns HDBSCAN calls 94.0% of sales noise and keeps two small, well-separated clusters, which the silhouette rewards. On data with no gaps, a density method has nothing to find, and the selection rule cannot tell.

3.4.5 Silhouette versus ground truth

On D6, DBSCAN has the best silhouette (0.42 against 0.28 for K-Means++) but a far worse ARI (0.31 against 0.90): it scores only the 35% of points it keeps and drops the other 65% as noise. On D5, HDBSCAN also has the top silhouette (0.228), computed on the 29% of points it does not call noise, and an ARI of only 0.050. In both cases silhouette excludes noise, so a method can raise it by discarding the hard points. [**All**: One paragraph, one line per dataset: D4 under the rule DBSCAN has the top silhouette (0.560) but ARI 0.039, against 0.731 for K-Means++; Spearman −0.80 (Ky); D7 K-Means top silhouette 0.343, GMM best ARI 0.774 vs 0.654 (Hoang Anh); D6

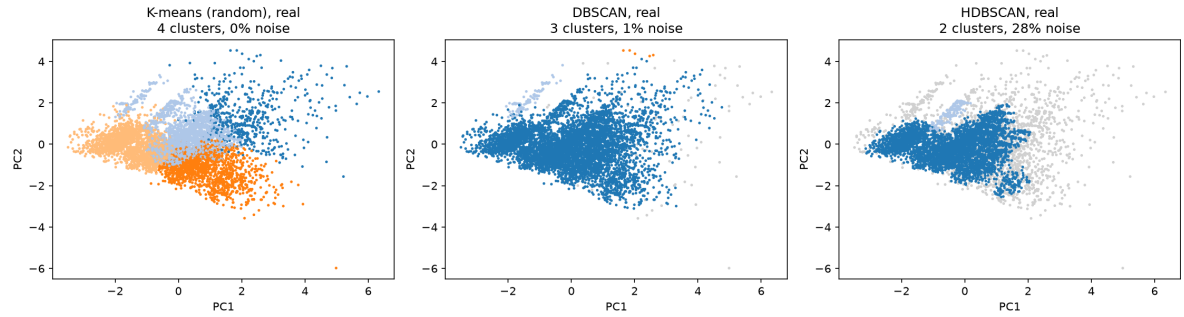


Figure 3: D8 at the chosen settings on recorded data, drawn on the first two principal components (6,000 sales; grey: noise). K-means cuts the mass into four segments; DBSCAN and HDBSCAN return one dominant cluster. Cause: no gaps between groups.

DBSCAN silhouette 0.42 vs K-Means++ 0.28, ARI 0.31 vs 0.90 (Lam); D8 unscaled K-means silhouette 0.540 from price bands, against 0.317 standardised (Wei Yew, written in Ablation C); D2 K-means silhouette flat at 0.516 to 0.532 for every K from 4 to 12, so it cannot choose K (Wei Yew, `weiyew/sc4020-earthquake/results/kmeans_k_selection.csv`); D5 HDBSCAN top silhouette with 71% noise (Lam).]

D7 Breast cancer. Given $k = 2$, the four methods that assign every point score almost the same silhouette, 0.314 to 0.343, yet their ARI runs from 0.575 to 0.774 (Table 5). The two scores disagree at the top: K-means has the highest silhouette of the four and GMM the lowest, while GMM matches the diagnosis best. The silhouette rewards compact clusters of similar size. The malignant class is more spread out than the benign one (Section 3.1), and a GMM with a full covariance per component can fit that difference where K-means cannot. Agglomerative clustering with Ward linkage optimises the K-means objective greedily and does worst of the four (ARI 0.575). The density methods fail outright. None of the 24 settings tried gives at least 2 clusters with at most 50% noise. The least noisy two-cluster settings discard 55.5% (DBSCAN) and 59.2% (HDBSCAN) of the tumours, and on what remains they score the two highest silhouettes of any method, 0.356 and 0.364, with the two lowest ARI, 0.222 and 0.156. Without the noise cap of Section 3.3, the silhouette would rank the two worst results first.

D6 Wine and the fake-data check. D6 repeats the pattern inside the noise cap. DBSCAN and HDBSCAN pass the cap with about a third of the wines as noise, take the two highest silhouettes (0.356 and 0.352) and the two lowest ARI (0.296 and 0.342), while GMM and K-means reach 0.915 and 0.897 with silhouettes near 0.28. The fake-data check does not separate good from bad here either. Every method that assigns all points scores far above its shuffled-column silhouette on both datasets (0.285 against 0.066 for K-means on D6, 0.343 against 0.033 on D7), and the density methods form no cluster at all on shuffled data. So each result passes the check, including the density results that match the labels worst: what they find, a dense core of one class, is real structure, but it is not the class structure. The check shows that a result is not an artefact of the marginals. It does not show which of several real structures a method has found.

3.5 Ablation studies

3.5.1 A: Initialisation (K-Means vs K-Means++)

On D3 ($K = 5$) and D4 ($K = 7$) each initialisation ran from one start with 30 seeds. On D4, K-Means++ lowers the mean inertia by 25% (845 to 632) and the mean iterations from 17.7 to 11.0, and raises ARI from 0.485 ± 0.095 to 0.561 ± 0.104. That gain is a third of the gap between the two main runs (10 starts, seed 42: ARI 0.493 against 0.731), so the larger figure rests on one

Table 5: D6 and D7 in the shared columns. S: silhouette; DB: Davies–Bouldin, both on clustered points only; null S: silhouette of the same setting on shuffled columns ("none": no clusters formed). Every fit takes under 0.3 s. On D7 the two density rows break the 50% noise cap and are shown for comparison only. Bold: best S and best ARI per dataset.

Data	Method	Clusters	Noise	Largest	S	DB	ARI	NMI	Null S
D6	K-Means ($k = 3$)	3	0%	36.5%	0.285	1.389	0.897	0.876	0.066
D6	Agglomerative (Ward, $k = 3$)	3	0%	36.0%	0.277	1.419	0.790	0.786	0.045
D6	Spectral ($k = 3$)	3	0%	36.0%	0.283	1.391	0.880	0.861	0.038
D6	GMM (3, diagonal)	3	0%	39.3%	0.280	1.412	**0.915**	0.892	0.054
D6	DBSCAN (2.33, 10)	2	33.7%	49.4%	**0.356**	1.062	0.296	0.391	none
D6	HDBSCAN (5)	2	32.6%	48.3%	0.352	1.093	0.342	0.430	none
D7	K-Means ($k = 2$)	2	0%	65.9%	0.343	1.321	0.654	0.532	0.033
D7	Agglomerative (Ward, $k = 2$)	2	0%	67.7%	0.339	1.370	0.575	0.457	0.018
D7	Spectral ($k = 2$)	2	0%	67.0%	0.337	1.316	0.761	0.663	0.023
D7	GMM (2, full)	2	0%	62.4%	0.314	1.377	**0.774**	0.661	0.043
D7	DBSCAN (2.33, 10)	2	55.5%	42.0%	0.356	0.853	0.222	0.255	none
D7	HDBSCAN (5)	2	59.2%	38.5%	**0.364**	0.844	0.156	0.213	none

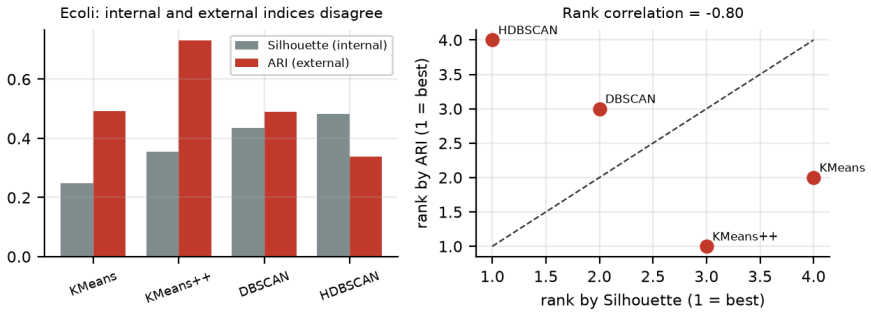


Figure 4: D4, silhouette against ARI for the four methods at each method's best-ARI setting. Left: the two scores. Right: their ranks; points on the dashed line would mean the two agree. The silhouette ranks the methods almost in reverse (Spearman $\rho = -0.80$); the same holds at the settings chosen without labels (Table 4). Cause: the silhouette ignores noise and favours compact clusters.

seed. On D3 the mean inertia is the same for both (within 0.01%); only the iterations fall, from 22.7 to 15.5, and two runs agree at ARI 0.96 with random starts and 0.98 with K-Means++.

Table 6 measures each run the same way as the D2 and D8 runs below: how far its inertia lies above the best of all 60 runs. K-Means++ moves most D4 starts close to the best (mean gap 8.9% against 45.7%), but its worst start is as bad as the worst random one (55%). Figure 5 shows why the ARI gain is modest. The starts fall into two groups, one near ARI 0.73 and one near 0.5, and K-Means++ reaches the high group in 8 of 30 seeds against 3 of 30. It raises the chance of a good start, but one start is still not enough. On D3, 29 of 30 starts with either method reach the best inertia, and one lands 28% above it.

Table 6: Ablation A on D3 and D4: 30 seeds, one start each. Gap: inertia above the best of all 60 runs. Agreement: mean ARI between pairs of runs.

Data	Start	Mean gap	Worst gap	Iterations	ARI	Agreement
D3 ($K = 5$)	random	0.9%	28.2%	22.7	–	0.961
D3 ($K = 5$)	K-Means++	0.9%	28.4%	15.5	–	0.979
D4 ($K = 7$)	random	45.7%	55.8%	17.7	0.485 ± 0.095	0.645
D4 ($K = 7$)	K-Means++	8.9%	55.5%	11.0	0.561 ± 0.104	0.678

On D2 and D8 each initialisation ran from one start with 20 seeds at the chosen K; we measure how far each run's inertia lies above the best run. On D2 ($K = 6$) K-Means++ helps clearly: the

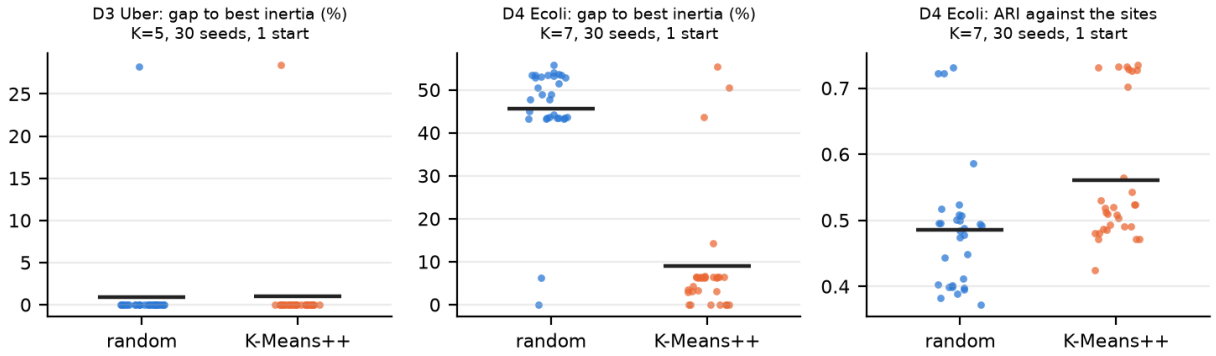


Figure 5: Ablation A, one dot per seed (30 per start method); bars mark the mean. Left and centre: inertia above the best run on D3 and D4. Right: ARI on D4. K-Means++ moves most D4 starts near the best inertia, and reaches the high-ARI group (about 0.73) in 8 of 30 seeds against 3 of 30 for random starts. Cause: spreading the starting centres out makes a good local optimum more likely, not certain.

mean gap falls from 10.9% to 4.8%, the worst from 18.8% to 11.5%, and iterations from 15.3 to 10.3. It does not remove the need for restarts. Two K-Means++ runs still agree only at ARI 0.80 on average, against 0.78 for random starts, because the belts give many local optima of similar inertia. On D8 ($K = 4$) the choice does not matter: all 40 runs end within 0.009% of the best inertia. Initialisation matters when the objective has many local optima of different quality; on D8 every start reaches practically the same one.

3.5.2 B: Distance on geographic coordinates

Every method ran at its chosen setting twice on D2: once on the sphere (Section 3.2) and once on latitude and longitude as flat numbers (DBSCAN's ε converted at 111.2 km per degree). To isolate the effect of the 180° meridian we count *cross-line pairs*: the 92 pairs of earthquakes within 50 km of each other on opposite sides of the line. Table 7 shows that flat coordinates separate every one of these pairs, for every method (Figure 6). The Fiji–Tonga–Kermadec system is cut in two. K-Means++ changes everywhere: its silhouette falls from 0.528 to 0.404 and its partition agrees with the sphere partition at ARI 0.557 only. DBSCAN hides the failure in its global numbers. Its silhouette moves from 0.703 to 0.701 and its partition still agrees at ARI 0.948, because only the clusters that cross the line are split. A check aimed at the known problem, here the cross-line count, finds what the internal scores miss.

Table 7: D2 on the sphere and on flat degrees. Cross-line pairs: 92 close pairs that straddle the 180° meridian; "kept" counts pairs placed in the same cluster.

Method	Distance	Clusters	Silhouette	ARI vs sphere	Pairs kept
K-Means++	sphere	6	0.528	1.000	92 of 92
	flat	6	0.404	0.557	0 of 92
DBSCAN	sphere	41	0.703	1.000	86 of 92
	flat	43	0.701	0.948	0 of 92
HDBSCAN	sphere	28	0.757	1.000	79 of 92
	flat	23	0.721	0.784	0 of 92

D3 lies more than 105° from the 180° meridian and spans 0.45° of longitude, so the cut cannot occur there; across its 0.35° of latitude the projection to kilometres changes the east–west scale by at most 0.3%.

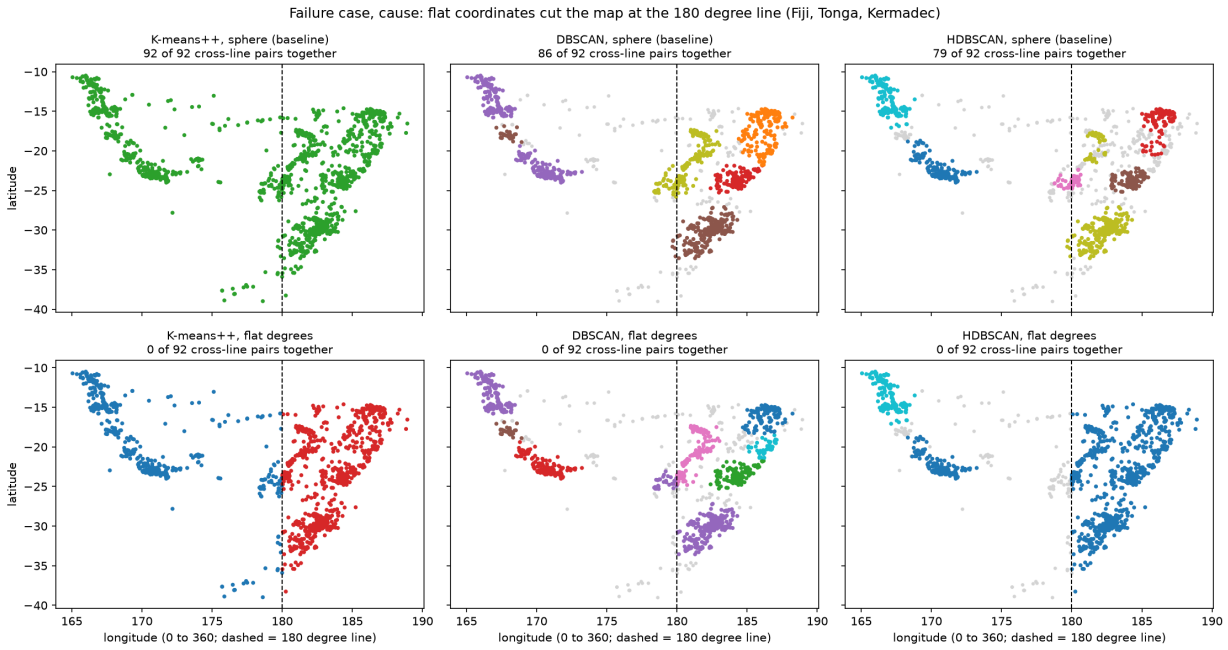


Figure 6: Fiji, Tonga and Kermadec, drawn on longitudes 0 to 360° (dashed: the 180° meridian). Top: on the sphere, clusters cross the line. Bottom: on flat degrees, every method splits the system at the line. Cause: flat coordinates treat 179.9°E and 179.9°W as 359.8° apart.

3.5.3 C: Feature scaling

On D4, standardising the seven features raises K-Means++ ARI ($K = 7$) from 0.497 to 0.731 (Figure 7, right). DBSCAN, given its best ε in each case, moves slightly the other way, from 0.481 to 0.455. K-means sums squared distances, so until the features are scaled the ones with the widest range decide the partition. DBSCAN only counts neighbours within ε, and re-tuning ε absorbs most of a change of scale.

On D8, K-means ($K = 4$) on raw units scores a silhouette of 0.540, well above the 0.317 of the standardised run. The higher score is not a better clustering. In dollars, price makes up 100.0% of the squared distance between sales, so the four clusters are price bands and floor area, lease and storey play no part. The raw partition agrees with the standardised one at ARI 0.261 only, and neither matches flat type (ARI 0.229 raw, 0.214 standardised). Scaling decides which question the clustering answers. The silhouette, measured in each run's own units, cannot say which is right.

3.5.4 D: Dimensionality

We project D4 onto its first 2, 3, 4 and 5 principal components, keep all 7 features as the last case, and tune both methods again in each: DBSCAN with `minPts` 5 and the ε with the best ARI, K-Means++ with $K = 7$ (Table 8). DBSCAN leads at 2 and 3 components, and K-Means++ from 4 on. DBSCAN peaks at 3 components and then falls, while K-Means++ rises from 4 components to 7, where it leads by 0.276. As components are added, distances between points become more alike [3], so a radius separates dense from sparse less well, while the added components still carry class information that K-means can use. The viable-ε column shows a second effect. On 2 components only 12.5% of the ε grid gives at least 2 clusters with under 50% noise, so DBSCAN's best result there would be hard to find without labels. We vary the number of PCA components, not the intrinsic dimension of the data, which we do not know.

Table 8: Ablation D on D4: ARI against the number of principal components. Viable ε: share of a 40-point ε grid that gives at least 2 clusters with under 50% noise.

Components	DBSCAN ARI	K-Means++ ARI	Viable ε	Better
2	**0.575**	0.468	12.5%	DBSCAN
3	**0.618**	0.466	67.5%	DBSCAN
4	0.453	**0.512**	57.5%	K-Means++
5	0.399	**0.520**	67.5%	K-Means++
7 (all)	0.455	**0.731**	65.0%	K-Means++

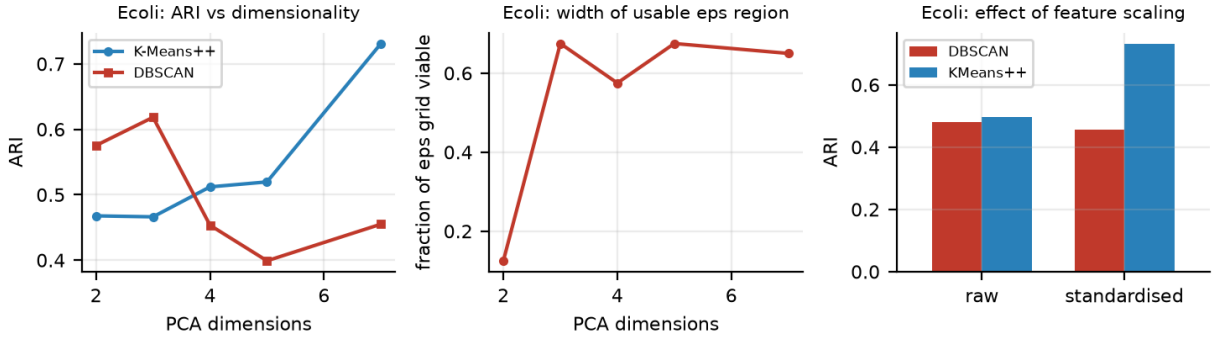


Figure 7: Ablations C and D on D4. Left: ARI against the number of principal components; DBSCAN leads at 2 and 3, K-Means++ from 4. Centre: share of the ε grid that gives at least 2 clusters with under 50% noise. Right: standardisation helps K-Means++ (0.497 to 0.731) and not DBSCAN (0.481 to 0.455). Cause: K-means depends on feature scale, and a density threshold weakens as dimensions are added.

3.5.5 E: Values recorded in steps

On D8 the selection rule picks density settings dominated by one cluster: it holds 97.1% of sales for DBSCAN, and 68.6% for HDBSCAN, which calls another 28.5% noise (Section 3.4.4). A natural repair is to add a cap on the largest cluster. We tested caps from 90% down to 50% on the data as recorded and on jittered data (Section 3.2). On recorded data a cap of 50% does give balanced partitions, but of the wrong kind: all 27 DBSCAN clusters, and 6 of the 7 HDBSCAN clusters, each contain a single storey band (Figure 8). They are stacks of identical recorded values, not groups of similar flats. On jittered data, where those ties are spread out, *no* setting meets a cap of 50% or 60%. The balanced partitions existed only because of the recording steps. At the uncapped HDBSCAN setting, by contrast, recorded and jittered data agree (ARI 0.886), so the chosen setting is not a tie artefact. Two lessons follow. Stepped values must be checked before a density method runs. And a constraint added to a selection rule can make it pick artefacts, which is why we kept the rule unchanged.

3.5.6 F: Data subsets

Each D2 subset is compared with a random subset of the same size, so that an effect of the subset is not confused with an effect of size. Clustering January to June or July to December separately, K-Means++ reproduces its full-year labels at ARI 0.963 and 0.829 (random halves: 0.968 and 0.994); the drop for the second half comes from aftershock sequences. Density methods change as much on random halves as on the real ones (HDBSCAN 0.414 and 0.472, against 0.505 and 0.544). Their parameters are counts of points, so their output depends on how many points there are. Keeping only magnitude 5.0 and above (1,780 earthquakes) behaves like a random subset of the same size (DBSCAN ARI 0.343 against 0.315, HDBSCAN 0.322 against 0.303). Larger earthquakes do not outline the belts more clearly, so we keep the 4.5 threshold.

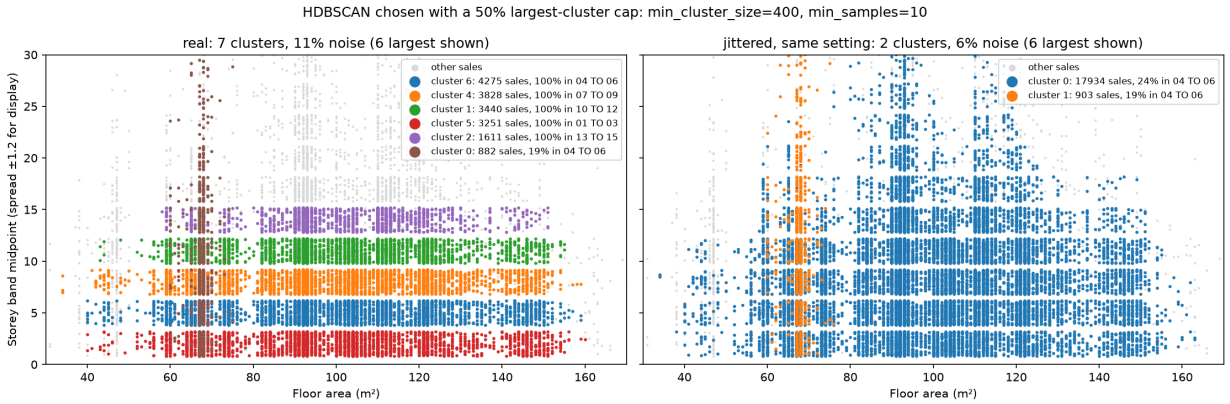


Figure 8: D8, HDBSCAN chosen under a 50% largest-cluster cap. Left, recorded values: 6 of 7 clusters are single storey bands. Right, the same setting on jittered values: 2 clusters. Cause: storeys recorded in 3-storey bands create exact ties that density methods read as dense regions.

3.6 Improvements to the methods

Table 9: Implemented improvements. Each fix targets a failure observed in Section 3.4.

Failure	Cause	Fix	Data	Before → after
Clusters split at 180°	flat lat/lon	surface distance	D2	0 → 92 of 92 pairs kept
Poor local optimum	random seeding	K-Means++	D4	ARI 0.485 → 0.561 (30 seeds)
Sparse belts lost	one global ε	HDBSCAN	D2	Americas kept 20% → 62%
One feature dominates	unequal scales	standardisation	D4	ARI 0.497 → 0.731
Clusters are storey bands	recorded steps	jitter within step	D8	band clusters 107 of 124 → 0 of 23
Curved clusters cut	convex assumption	spectral clustering	D1	ARI 0.481 → 1.000
Classes of unequal spread	equal-size assumption	GMM, full covariance	D7	ARI 0.654 → 0.774
Degenerate k chosen	silhouette alone	constraints + fake-data check	D2, D3	D2 null gap 0.015 → 0.144; D3 0.089 → 0.113

Surface distance (D2). Haversine distance for density methods and 3D unit vectors for K-means measure what matters, the distance along the Earth's surface. The fix is exact, and its only cost is that ε must be given in kilometres and the coordinates converted. It changes the answer for every method (Ablation B), which makes it a correctness fix rather than a tuning choice.

HDBSCAN instead of DBSCAN (D2). HDBSCAN replaces the single radius with a hierarchy of densities and keeps the clusters that persist longest, so sparse and dense belts can both survive. In the Americas it clusters 62% of earthquakes against 20%. The costs are real: dense cores are trimmed (78% against 90% in the south-west Pacific), each fit takes about 20 times longer (1.08 s against 0.05 s), and `min_cluster_size` is a count, so the result changes with the size of the dataset (Ablation F).

Jitter within the recording step (D8). Moving each stepped value by up to half its step spreads exact ties without moving any sale into another band. Across the 9 HDBSCAN settings, 107 of the 124 clusters found on recorded data are single storey bands; on jittered data, 0 of 23 are. The fix costs nothing at the chosen setting (recorded and jittered agree at ARI 0.886), but it adds randomness, so the seed must be fixed and reported. It also shows that D8 has fewer real density clusters than the recorded data suggest.

Fake-data check for choosing K (D2). The silhouette alone picks $K = 2$ on D2 (0.544, the highest of any K), but shuffled coordinates score 0.529 at the same K: the split follows the distribution of latitude and longitude on their own, not their combination. Requiring a clear gap over the fake data rules out $K = 2$ and $K = 3$. From $K = 4$ the gap is 0.126 to 0.149 and almost

flat, so the final choice, $K = 6$, rests on agreement between seeds (ARI 0.993). The check costs extra clustering runs on fake data for each K.

GMM instead of K-means (D7). K-means cuts the space halfway between two centres, which suits clusters of equal spread. On D7 the malignant class is the wider one, so the boundary falls inside it: 36 of the 212 malignant tumours join the benign cluster. With GMM that falls to 16. A GMM with one full covariance matrix per component models the spread of each class and moves the boundary accordingly: ARI rises from 0.654 to 0.774 and NMI from 0.532 to 0.661. The costs are a slower fit, 465 covariance parameters per component estimated from 569 tumours, and a lower silhouette (0.314 against 0.343), so an analyst who trusts the silhouette would reject the better result. BIC, the model's own criterion, also selects two components with full covariance, so this fix can be chosen without labels.

Proposed, not implemented (D6, D7). The density methods fail on both tables because 13 and 30 standardised features leave little contrast in density. Projecting onto a few principal components before DBSCAN or HDBSCAN is the natural next step; on D7 the first two components already hold 63.2% of the variance.

Spectral clustering for curved clusters (D1). K-Means++ fails on D1 because it assigns each point to the nearest centroid, so cluster boundaries are straight lines. Spectral clustering replaces Euclidean distance with connectivity in a 10-nearest-neighbour graph: points along one moon are linked by short hops, the two moons are not, and the eigenvectors of the graph Laplacian separate them. ARI rises from 0.481 to 1.000. The cost is that k and the graph must still be chosen, and the affinity matrix and eigenvectors limit how far the method scales (Section 3.7).

Proposed, not implemented (D2). D2 could be scored rather than inspected: assigning each earthquake to the nearest segment of a published plate-boundary model would give labels for ARI, and show directly whether a cluster follows one boundary or joins several. OPTICS [1], which orders points by reachability so that clusters at several densities can be read from one plot, would show whether a single ordering explains both the dense and the sparse belts. [**Lam, Ky**: other proposals, e.g. approximate spectral clustering for large n.]

3.7 Runtime and scalability

D8 is large enough to measure how each method grows. Each method ran at its chosen setting on random subsets of 1,000 to 234,294 sales (K-means from one start). Figure 9 plots the median time against size on log scales, where the slope of a line estimates the exponent b in $t \propto n^b$. K-means is far cheaper than either density method at every size: under 0.06 s on all 234,294 sales. Its fitted slopes (0.57 and 0.67) are below 1 only because a fixed start-up cost dominates at these sizes; each iteration is linear in n. HDBSCAN grows with slope 1.79, from 0.02 s at 1,000 sales to 212 s at 234,294. DBSCAN grows with slope 1.65, but at full size it was not run at all, because of memory rather than time. With a fixed ε, every sale has more neighbours as the data grow. Estimated from 2,000 probe points, the full data would hold about 742 million neighbour pairs, about 6 GB for the neighbour indices alone, which scikit-learn's DBSCAN stores in full. HDBSCAN, slower at every size, did run on all sales within 16 GB.

D5 was sampled down from 70,000 to 3,000 images because three methods scale badly with n in 50 dimensions. Spectral clustering needs an $n \times n$ affinity matrix: dense, at $n = 70{,}000$ it takes about 39 GB in double precision, and the eigenvector step grows faster than linearly even with a sparse graph. DBSCAN and HDBSCAN rely on neighbour queries, and in 50 dimensions spatial indexes prune little, so these queries approach the $O(n^2)$ brute-force cost. On the 3,000-point sample one fit takes 0.011 s for DBSCAN, 0.079 s for K-Means++, 0.18 s for Spectral, 0.30 s for

HDBSCAN and 2.80 s for GMM. GMM is the slowest here: with full covariances each EM step costs $O(nkd^2)$ with $d = 50$. These times exclude parameter search, which refits DBSCAN 15 times and HDBSCAN 6 times. On D3 (15,000 points, 2 dimensions) one fit takes 0.04 to 0.06 s for K-means, 0.52 s for DBSCAN and 1.1 s for HDBSCAN; on D4 (336 proteins) every method takes under 0.03 s (Table 4). At these sizes runtime does not decide the choice of method.

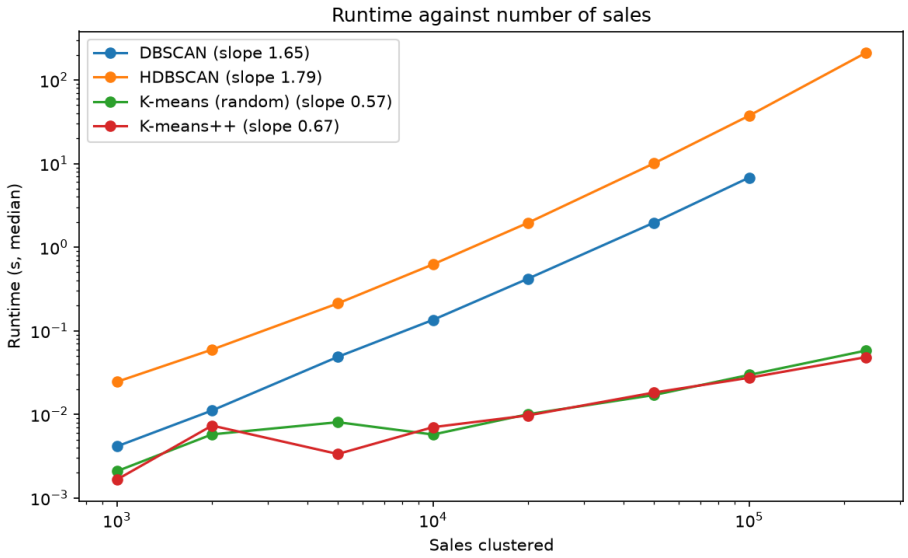


Figure 9: D8, median fit time against the number of sales, log–log, at the chosen settings; the slope of each line is in the legend. DBSCAN stops at 100,000 sales: the full data would need about 6 GB of neighbour indices.

3.8 Success and failure cases

3.8.1 Successes

HDBSCAN on D2: belts of varying density. Shown in Figure 2. HDBSCAN keeps the sparse belts of the Americas (62% of earthquakes clustered, against 20% for DBSCAN) as long clusters that follow the plate boundary. It succeeds because its density threshold adapts to each cluster, the property D2 needs most.

GMM on D7: classes of unequal spread. Shown in Figure 12. GMM matches the diagnosis best of all methods (ARI 0.774). K-means places its boundary too far into the wide malignant class (ARI 0.654); the full covariance of each GMM component lets the boundary follow the compact benign class. Cause of success: the model allows clusters of different spread.

Spectral and DBSCAN on D1. Both follow the curve of each moon (Figure 10): Spectral reaches ARI 1.000 and DBSCAN 0.982, leaving 9 outlying points as noise. They succeed for the same reason: each moon is a chain of close neighbours, and the gap between moons is wider than any link in the chain. K-Means++ and GMM fail on the same data because they assume convex clusters.

K-Means++ on D4: compact classes in seven dimensions. With $K = 7$, K-Means++ matches the localisation sites best of all methods on D4 (ARI 0.731; with the $K = 5$ the silhouette picks, 0.697). It keeps 96% of the largest class (cytoplasm, 143 proteins) in one cluster, 86 to 91% of the three mid-sized classes, and all 5 proteins of the outer-membrane lipoproteins together. It succeeds because the classes are compact groups in seven dimensions that touch, with no density

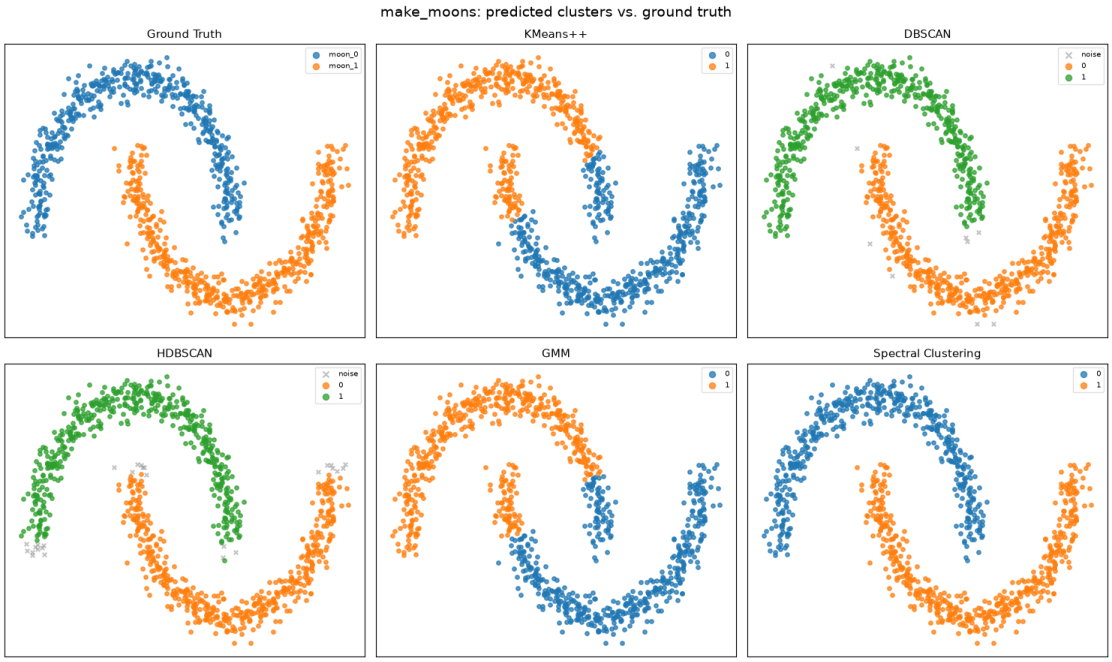


Figure 10: D1 make_moons. K-Means++ and GMM cut both moons with a straight boundary (ARI 0.48, 0.49); DBSCAN, HDBSCAN and Spectral follow the curves (ARI 0.98, 0.95, 1.00). Grey crosses: noise.

gap between them: the case K-means is built for, and the case where a density threshold fails (Ablation D). The inner-membrane proteins (77) are its weak spot: they overlap the related inner-membrane classes, and only 52% end up in one cluster.

HDBSCAN on D8, floor area and lease only: flat designs. This feature set was chosen after seeing the main results, so we present it as an ablation finding and a sanity check, not as a tuned setting. Dropped to two features, D8 does contain dense spots: standard flat designs, built to fixed plans in particular years. HDBSCAN finds 14 clusters (43.2% noise), and 13 of them are at least 97% one flat type (Figure 11); in the median cluster one flat model accounts for 83% of sales. A simple grouping by flat model and lease would give similar groups, which is the point: the method recovers a structure we can name. ARI against flat type is still only 0.127, because each type is split into several designs and the noise counts as a cluster. A low ARI here reflects a finer partition than the labels, not a failure.

3.8.2 Failures

K-Means++ on D2: unrelated belts merged. Shown in Figure 1. One cluster joins five belts whose centres are 2,932 to 11,449 km apart, from Iceland to the Mid-Indian Ridge. Cause: cluster shape. Compact regions cannot follow long belts.

Density methods on D8: one mass, or storey bands. Shown in Figures 3 and 8. At the chosen settings DBSCAN puts 97.1% of sales in one cluster. Under a cap on the largest cluster, the clusters become single storey bands. Causes: no gaps between groups, and values recorded in steps.

DBSCAN on D7: half the data discarded. Shown in Figure 12. The least noisy two-cluster setting keeps the dense benign core as one cluster, finds a second cluster of 14 malignant tumours and labels 55.5% of tumours noise, including 183 of the 212 malignant ones. Its silhouette (0.356) is higher than that of GMM or K-means, and its ARI is 0.222. Causes: 30 dimensions, and one class much sparser than the other.

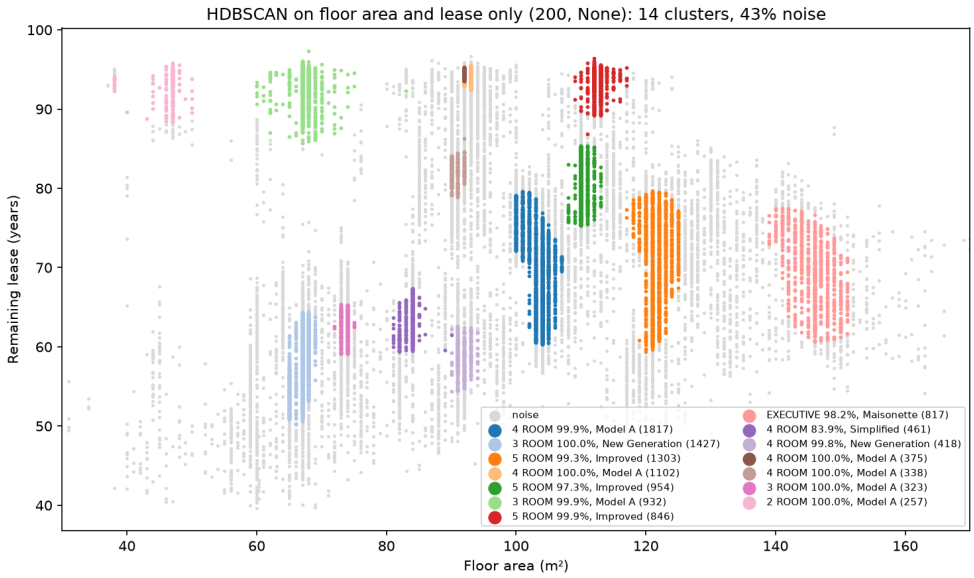


Figure 11: D8, success of HDBSCAN on floor area and remaining lease only (`min_cluster_size` 200; grey: noise). Each cluster is one standard flat design; the legend gives its main flat type and model. Cause of success: fixed building plans create real dense spots in these two features.

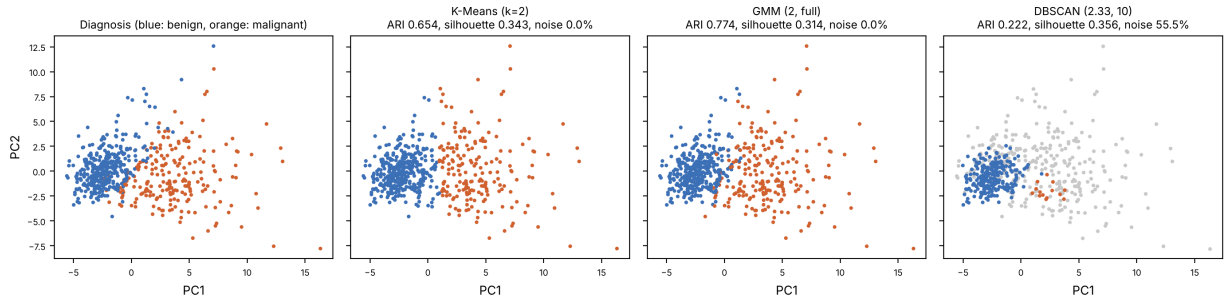


Figure 12: D7 on its first two principal components (grey: noise; each cluster takes the colour of its majority class). Success of GMM against K-means: the boundary follows the compact benign class. Failure of DBSCAN: only the dense benign core is clustered. Causes: classes of unequal spread, and no density contrast in 30 dimensions.

DBSCAN on D5. DBSCAN puts most of the ten classes into one cluster, a second small cluster at the edge, and 33% of images into noise (ARI 0.033; Figure 13). The cause is one global ε in 50 dimensions: an ε large enough to connect points within a class also connects the overlapping classes to each other, and a smaller one turns most points into noise. The k-based methods, which are not tied to a density threshold, reach ARI 0.34–0.40 on the same data.

HDBSCAN on D4: rare classes called noise. Even at its best setting by ARI, a minimum cluster size of 12, HDBSCAN returns two clusters and calls 37% of proteins noise (ARI 0.339), yet it has the highest silhouette of the four methods at their best-ARI settings (0.483, Figure 4). The noise is not spread evenly. It takes 95% of the outer-membrane class (20 proteins), every protein of two of the three smallest classes (5 and 2), and about half of the inner-membrane and periplasm classes. Cause: a class of 2 to 20 proteins in seven dimensions is not dense enough to form a cluster of 12, so HDBSCAN discards exactly the classes that are hardest to find, and the silhouette, which skips noise, rewards it. The setting the rule picks without labels avoids the noise only by putting 97% of proteins in one cluster (ARI 0.038).



Figure 13: D5 Fashion-MNIST (3,000 images; clustered in 50 PCA dimensions, plotted in the first two). DBSCAN merges most classes into one cluster (ARI 0.03); HDBSCAN calls 71% of points noise (ARI 0.05). Spectral, GMM and K-Means++ reach ARI 0.40, 0.39, 0.34.

All: success/failure panel grid: one panel per case above

Figure 14: [**All**: caption naming the cause of each case]

3.9 Key factors and practical guidance

Key factors. Ordered by the size of their effect in our experiments:

1. **Cluster shape against the method's assumption.** On D1 K-Means++ reaches ARI 0.481 and DBSCAN 0.98; on D2 K-means joins belts thousands of kilometres apart (Section 3.4.1). No parameter setting repairs a wrong assumption about shape.
2. **Gaps between groups.** Density methods need empty space between clusters. Where there is none they return one dominant cluster: 97.1% of sales on D8, 96.1% of pickups on D3 and 97% of proteins on D4 at the settings chosen without labels (Sections 3.4.4 and 3.4.2).
3. **Dimensionality.** On D4 the ranking of DBSCAN and K-Means++ reverses between 3 and 4 principal components (Ablation D); on D5, with 50 dimensions, DBSCAN reaches ARI 0.03 and HDBSCAN calls 71% of points noise (Section 3.4.3).
4. **Uneven density.** One DBSCAN radius clusters 90% of the dense south-west Pacific but 20% of the Americas on D2; HDBSCAN raises the Americas to 62% (Section 3.4.2).
5. **Parameter choice without labels.** On D4 the rule's DBSCAN setting scores ARI 0.039 against 0.489 at the best setting; on D3 the knee and the silhouette-best radius differ threefold and give 13 and 2 clusters (Table 4).
6. **Feature scaling.** Standardising D4 raises K-Means++ ARI from 0.497 to 0.731; on D8, raw units turn K-means into price bands (Ablation C).

7. **Values recorded in steps.** On D8, 107 of 124 HDBSCAN clusters are single storey bands on the recorded data and none on jittered data (Ablation E).
8. **Initialisation.** Over 30 seeds K-Means++ raises mean ARI on D4 from 0.485 to 0.561, and on D2 it cuts the mean gap to the best inertia from 10.9% to 4.8%; on D3 and D8 it makes no difference to the result (Ablation A).
9. **Runtime.** It only matters at the size of D8, where HDBSCAN takes 212 s and DBSCAN runs out of memory (Section 3.7).

Practical guidance.

1. **Choose by the structure of the data, not by a score.** Plot the data or a projection first. If the groups are curved or long, no choice of k will help K-means; if they are compact and touching, or the data have more than a handful of dimensions, no choice of ε will help DBSCAN.
2. **Do not pick settings by the silhouette alone.** It favours few, round clusters and settings that call hard points noise. First restrict the search to what the analysis needs (a noise cap, a minimum number of clusters), then compare within that range.
3. **Report the noise fraction and the largest cluster next to every silhouette.** A silhouette computed after discarding 37% of the points, or with 97% in one cluster, says little.
4. **Compare every result against fake data.** Rerun the chosen setting on shuffled columns or uniform points. It catches results with no real structure (D4 density methods, D8), though not results with the wrong structure (D2).
5. **Check for values recorded in steps.** Tied values such as storey bands form dense spots that density methods report as clusters; add jitter within each step and see whether the clusters remain.
6. **Standardise before K-means, and use K-means++ with several starts.** One K-means++ start is not enough when the objective has many local optima (D2).
7. **Prefer a parameter whose unit means something, and measure distance correctly.** A radius in kilometres can be checked by someone who knows the area; a k cannot. Use distance on the sphere for global coordinates (D2); a local projection is enough within a city (D3).

4 Conclusion

We compared K-means, K-means++, DBSCAN and HDBSCAN on eight datasets, each chosen to test one data property, with GMM, spectral and agglomerative clustering as reference points. Density beats centroids when clusters are separated by gaps and have any shape, in few dimensions: on D1 DBSCAN reaches ARI 0.98 against 0.48 for K-means, and on D2 HDBSCAN follows belts that K-means merges. Centroids win when classes are compact and touch, or the data have many dimensions: on D4 K-Means++ reaches ARI 0.697 without labels while both density methods fall to 0.04, and the ranking reverses between 3 and 4 principal components. Where the data have no gaps at all, as in the core of D3 and in D8, neither family finds natural groups. Density methods return one dominant cluster, and K-means draws borders where its objective puts them.

The silhouette did not tell us which case we were in. On five of the six labelled datasets the method with the top silhouette was not the one that matched the labels best, and on D4 it ranks the four methods almost in reverse (Spearman $\rho = -0.80$). Rerunning each result on fake data helps where a method finds nothing real (D4, D8), but not where it finds the wrong structure (D2). Known remedies, namely surface distance, HDBSCAN for uneven density, standardisation

and K-means++, repaired the failures they target, each at a cost, and none produced clusters in data with no gaps. The choice of method should follow from what the data look like and what the analysis needs, and every score should be checked against fake data and against the points it ignores.

Limitations

On D6 and D7, the methods that need k were given the number of classes, which an analyst would not know. The label-free criteria happen to agree here (the silhouette picks $k = 3$ on D6 and $k = 2$ on D7 for K-means), so the cost is small on these two tables, but both are small (178 and 569 points) and their density settings were ranked by DBCV, not by the silhouette used elsewhere.

D3 has no labels, so its clusters are judged against known geography only, and it is one 15,000-point sample of one month. The D4 settings with the best ARI use the labels and overstate what an analyst could reach without them; D4 is also small (336 proteins), so a single protein moves a rare class's share by up to 50%.

D2 has no labels, so its failures are judged against plate-boundary geography and cannot be scored. Its density parameters are counts of points, so the clusters change with the size of the catalogue: a random half of the year agrees with the full year at ARI 0.51 to 0.55 only for the density methods (Ablation F). D8 has no location beyond the town, so its clusters describe flats rather than places, and its one label, flat type, is nearly a function of floor area (ARI 0.601 from floor area alone), so a low ARI on D8 says little about the quality of a clustering.

Our evidence on dimensionality comes from two datasets, D4 with 7 features and D5 reduced to 50 principal components from 784 pixels, which keep 86.5% of the variance. We did not test the methods on the full 784 dimensions or beyond, so the high-dimensional results describe data that have already been reduced.

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